

Xiang Xiang Wang

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Research Interests: Mathematical AI and data science; geometric and topological data analysis; Grassmannian and Riemannian geometry; matrix analysis; machine learning for biological and imaging data.

RESEARCH EXPERIENCE

Postdoc Research Associate, University of Georgia

2026-present

Advisor: Guo-Wei Wei, University of Georgia

I develop mathematical AI methods for data science by integrating geometric and topological representations with machine learning. My research uses Grassmannian geometry and topological data analysis to capture structure in high-dimensional biological data. Recent projects include GrassTop for viral classification and phylogenetic analysis and data-adaptive Grassmann manifold representations for spatial transcriptomics alignment.

Research Associate, Michigan State University

2025-2026

Advisor: Guo-Wei Wei, Michigan State University

Developed geometric, topological, and spectral methods for biological and imaging data analysis, including Grassmann manifold models and sheaf-based methods. Designed mathematical representations and computational approaches for single-cell data analysis and image analysis, connecting mathematical modeling with data-driven applications.

Ph.D. Student in Mathematics, University of Nevada, Reno

2021-2025

Ph.D. Advisor: Tin-Yau Tam, University of Nevada, Reno

Investigated Grassmannian geometry and quaternion matrix methods for data representation, with applications to image and video analysis. Designed, implemented, and evaluated machine learning models for color image set recognition. Collaborated with researchers at the University of Nevada, Reno, Shanghai University, and Auburn University.

Statistical Analyst, Provost Office of University of Nevada, Reno

June 2023-August 2023

Designed data processing workflows for the Co-Requisite Assessment Program, including data cleaning, transformation, and statistical modeling using SPSS, R, and Python. Developed dashboards and reports to support program assessment and institutional decision-making.

Master's Student in Statistics, Shanghai University

2017-2020

Advisor: Qing-Wen Wang, Shanghai University

Completed graduate training in statistics, building a foundation for quantitative data analysis and computational research. Investigated quaternion matrices and eigenvalue problems, developed a fast algorithm for computing eigenvalues of large-scale quaternion matrices, and explored applications to image processing. This research led to a publication in the *Journal of Scientific Computing*.

TEACHING EXPERIENCE

Teaching Assistant, Michigan State University

Spring 2026

Course: MTH 133 Calculus II

Responsibilities: Overseeing instruction and evaluating quizzes

Instructor, Michigan State University

Fall 2025

Course: Mathematics-Assisted AI and AI-Inspired Mathematics

Responsibilities: Delivered four lectures on advanced topics including manifold theory and quaternionic Grassmannians as part of a multi-instructor course.

Instructor of Record, University of Nevada, Reno

Winter 2023, Summer 2023, Spring 2024

Course: Math 127 Precalculus II

Responsibilities: Developed and delivered two weekly 75-minutes lectures (over 100 students) and five weekly 3-hours lectures (20-30 students) in winter and summer semesters to non-majors from diverse backgrounds. Conducted instruction, set up assignments and exams, evaluated performance, and submitted final grades.

Instructor of Record, University of Nevada, Reno

Fall 2023

Course: Math 126E Precalculus I Expanded

Responsibilities: Designed and conducted five weekly 50-minutes lectures to new undergraduate students. Enhanced student engagement by implementing an encouragement and class bonus system.

Instructor of Record, University of Nevada, Reno

Summer 2023, Fall 2024, Spring 2025, Summer 2025

Course: Math 176 Introductory Calculus for Business and Social Sciences

Responsibilities: Delivered lectures focused on fundamental calculus concepts tailored to business students, emphasizing real-world applications in economics, finance, and social sciences. Designed assignments and exams that applied calculus to business scenarios, evaluated student performance, and provided comprehensive feedback.

Teaching Assistant, University of Nevada, Reno

Fall 2022, Spring 2023

Course: Math 126 Precalculus I

Responsibilities: Overseeing instruction and evaluating quizzes

Teaching Assistant, University of Nevada, Reno

Spring 2022

Course: Math 176 Introductory Calculus for Business and Social Sciences

Responsibilities: Overseeing instruction and evaluating quizzes

Grader, University of Nevada, Reno

Fall 2021, Fall 2023

Course: Theory of Positive Integers

Responsibilities: Evaluating assignments for Professor Raúl Rojas González, a distinguished faculty member known for his expertise in number theory.

PUBLICATIONS

Preprints and Manuscripts

- X.X. Wang, G.W. Wei, GrassTop: Grassmannian k-mer Topology for Viral Classification and Phylogenetic Analysis. Submitted to Theory in Biosciences, 2026. Preprint available on <https://arxiv.org/abs/2609.13341>
- X.X. Wang, S. Cottrell, G.W. Wei, Data-Adaptive Grassmann Manifold Representations for Spatial Transcriptomics Alignment. Submitted to SIAM Journal on Applied Mathematics, 2026. Preprint available on arXiv: <https://arxiv.org/abs/2609.12805>.
- S. Cottrell, X.X. Wang, G.W. Wei, A Review of Geometric, Topological, and Spectral Approaches in Single-Cell Data Analysis. Submitted to Current Opinion in Systems Biology, 2026.
- T.Y. Tam, X.X. Wang, Geometric Means and Their Properties of Grassmannians. Preprint available on arXiv: <https://arxiv.org/abs/2412.15161>, 2025

Published and Accepted

- X.X. Wang, G.W. Wei, A Hierarchical Sheaf Spectral Embedding Framework for Single-Cell RNA-seq Analysis. *SIAM Journal on Life Sciences*, accepted for publication, 2026. arXiv: [2603.26858](https://arxiv.org/abs/2603.26858).
- X.X. Wang, G.W. Wei, Multi-dimensional Persistent Sheaf Laplacians for Image Analysis. *Foundations of Data Science*, Early Access, 2026. DOI: [10.3934/fods.2026024](https://doi.org/10.3934/fods.2026024).
- X.X. Wang, S. Cottrell, G.W. Wei, Multiscale Grassmann Manifolds for Single-Cell Data Analysis, *Machine Learning: Science and Technology*, 7 (2026), 3 (20 pages)
- Z. Wang, D. Li, G. Wang, M. Zia, X. Wei, X.X. Wang, J. Jiang, Chaotic Dynamics-Regulated Topological Learning for Patient-Specific Preictal State Identification. Accepted for publication at *Machine Learning: Science and Technology*, 2026.
- X.X. Wang, T.Y. Tam, Quaternion Grassmannians for Color Image Set Recognition. *Journal of Mathematical Imaging and Vision*, 68 (2026), 13 (16 pages)
- Z.H. He, Y.Z. Xu, Q.W. Wang, X.X. Wang, Solving a system of quaternion matrix equations by using PSVD for multiple matrices with applications. *Journal of Computational and Applied Mathematics*, (2025), 117072 (18 pages)
- Z.H. He, T.T. Liu, X.X. Wang, Eigenvalues of Quaternion Tensors: Properties, Algorithms and Applications, *Advances in Applied Clifford Algebras*, 35 (2025), 4 (23 pages).
- Z.H. He, W.L. Qin, J. Tian, X.X. Wang, Y. Zhang, A new Sylvester-type quaternion matrix equation model for color image data transmission, *Computational and Applied Mathematics*. 43 (2024), 227 (30 pages)
- Z.H. He, X.X. Wang, Y.F. Zhao, Eigenvalues of Quaternion Tensors with Applications to Color Video Processing, *Journal of Scientific Computing*. 94 (2023), 1(15 pages)
- Z.H. He, C. Navasca, X.X. Wang, Decomposition for a quaternion tensor triplet with applications, *Advances in Applied Clifford Algebras*, 32 (2022), 9 (19 pages).
- Z.H. He, W.L. Qin, X.X. Wang, Some applications of a decomposition for five quaternion matrices in control system and color image processing, *Computational and Applied Mathematics*, 40 (2021), 205 (29 pages).
- S.W. Yu, Z.H. He, T.C. Qi, X.X. Wang, The equivalence canonical form of five quaternion matrices with applications to imaging and Sylvester-type equations, *Journal of Computational and Applied Mathematics*, 393 (2021), 113494 (20 pages).
- Z.H. He, C. Chen, X.X. Wang, A simultaneous decomposition for three quaternion tensors with applications in color video signal processing, *Analysis and Applications*, 19 (3) (2020) 529-549 (21 pages).
- Q.W. Wang, X.X. Wang, Arnoldi method for large quaternion right eigenvalue problem, *Journal of Scientific Computing*, 82 (2020), 58 (20 pages).
- G.H. Peng, X.X. Wang, Y.Z. Zhang, Multidimensional scaling analysis based on attribute reduction of bivariate mutual information, *International Mathematical Forum*, 12 (3) (2017) 111-118 (9 pages).

FUNDED GRANTS

Co-Principal Investigator, *Self-Assembly of Tunable Molecular Memristors with Long-Range Order for Resilient and Energy-Efficient Neuromorphic Computing*. U.S. Department of Energy, Office of Science, Basic Energy Sciences, through Pacific Northwest National Laboratory. \$360,000, June 15, 2026–January 14, 2029. PNNL Award No. PNNL0000883191; DOE FWP 84274. PI: Guo-Wei Wei.

PEDAGOGICAL TRAINING

Seminar in Teaching Mathematics and Statistics	Fall 2022
University of Nevada, Reno	
This course is designed to prepare graduate students for success as teaching assistants in mathematics and statistics. The activities, discussions, and work are designed to be useful and practical during the first semester of teaching and beyond.	
2024 NSHE Corequisite Conference	April 25-26, 2024
Western Nevada College Carson City, USA	
This conference provided a platform for educators to reflect on and celebrate their achievements, strengthen professional relationships, and expand their networks.	

EDUCATION

2021-2025	Ph.D. in Mathematics (Graduation Date: August 13, 2025) Department of Mathematics and Statistics, University of Nevada, Reno GPA: 4.00 (out of 4)
2017-2020	M.Sc. in Statistics Shanghai University Average Score: 90.25 (out of 100)
2013-2017	B.Sc. in Information and Computing Science North China University of Water Resources and Electric Power GPA: 4.14 (out of 4.5) Consistently ranked first in my major each semester

PRESENTATIONS

Oral Presentations	
- X.X. Wang, Grassmannian Methods for Single-Cell RNE- seq Data, The 27th Conference of the International Linear Algebra Society (ILAS 2026), Virgina, May 18-22, 2026. - X.X. Wang, Grassmannian Methods for Visual and Single-Cell Data, Joint Mathematics Meetings (JMM 2026), Washington, January 4-7, 2026. - X.X. Wang, Color Image and Video Recognition Based on Quaternionic Grassmannians, 2025 SIAM Great Lakes Section Annual Meeting, Chicago, September 27-28, 2025 - X.X. Wang, Quaternion Grassmannians for Color Image Set Recognition, Joint Mathematics Meetings (JMM 2025), Seattle, January 8-11, 2025. - X.X.Wang, Some Properties of Geodesic Triangles in Grassmannians, The 2024 Workshop Matrices and Operators (MAO 2024), Reno, USA, June 14-17, 2024. - X.X.Wang, Some Inequalities of Geometric Means in Grassmannians, Joint Mathematics Meetings (JMM 2024), San Francisco, January 3-6, 2024. - X.X. Wang, Geometric means and their properties of Grassmannians, The 10th International Conference on Matrix Analysis and Applications (ICMAA), Kunming, China, August 15-18, 2023. - X.X. Wang, Geometric means and their properties of Grassmannians, International Workshop on Matrix Analysis and Its Applications, Quynhon, Vietnam, July7-8, 2023. - X.X. Wang, Arnoldi method for right eigenvalue problem of the large-scale quaternion matrices, Joint	

Mathematics Meetings (JMM 2023), Boston, January 4-7, 2023.

- X.X. Wang, Arnoldi method for right eigenvalue problem of the large-scale quaternion matrices, Joint Mathematics Meetings (JMM 2022), Seattle, January 5-8, 2022, postponed due to COVID-19 and held virtually, April 6-9, 2022.

Poster presentations

- X.X. Wang, Some Inequalities of Geometric Means in Grassmannians, 6th Annual GSA Poster Symposium. 2023.

AWARDS

Research Grants

Graduate Student Association (GSA) Research Grant Award, University of Nevada, Reno. (\$1,000)	2021
Graduate Student Association (GSA) Travel Award, University of Nevada, Reno. (\$500)	2022

Fellowships

Graduate Dean's Merit Scholarship, University of Nevada, Reno. (\$10,000)	2021
National Scholarship, China. (CNY 8,000)	2015

Honors & Awards

Love of Learning Award, The Honor Society of Phi Kappa Phi.	2026
College of Science Outstanding Graduate Assistant Award, University of Nevada, Reno.	2025
College Graduate Excellence Award of Shanghai City, China.	2020
Excellent Student Award of Shanghai University, China.	2018
Third Prize in the 15th China Post-Graduate Mathematical Contest in Modeling, China.	2018
Outstanding Graduate Award of Henan Province, China.	2017
"Three Good Students" Award of General Higher Education in Henan Province.	2016
Second Prize in the Sixth National College Mathematics Competition, China.	2014

LEADERSHIP&COLLABORATION

- Organizer, AMS Special Session "Mathematical and Computational Methods for Biological Structures and Data," Spring Southeastern Sectional Meeting, Birmingham, AL, March 21, 2027 (upcoming).
- Co-organizer, special session at the Joint Mathematics Meetings (JMM 2027), upcoming.
- Organizer, MS34 "Mathematical and Computational Methods for High-Dimensional Biological Data," SIAM Texas-Louisiana Sectional Meeting, Dallas, October 30–November 1, 2026 (upcoming).
- Co-organizer, Special Session on Matrix Analysis and Its Applications, Joint Mathematics Meetings (JMM 2026), Washington, DC, January 2026.
- Chaired sessions at major conferences (JMM 2025, MAO 2024)

SERVICES

Moderator & Logistic Volunteer, 2025 Youth Security Forum: Telling Truth from Lies Misinformation,

Disinformation, AI Threats to National Security, UNR May 2025

Organizer, Lecture Series in Mathematical AI and Bioscience (MATH 8970), Department of Mathematics, University of Georgia. Fall 2026; Spring 2027 planned

Local Organizer, The 2024 Workshop Matrices and Operators (Mao 2024), UNR June 2024

Organizing Committee Member, Biweekly Matrix Seminar, Department of Mathematics and Statistics, University of Nevada, Reno. 2023–present

Mentor, Nevada GAIN Program

Journal Reviewer for

- Linear and Multilinear Algebra
- Advances in Operator Theory
- zbMath
- The Electronic Journal of Linear Algebra
- Numerical Algorithms
- IEEE Transactions on Signal and Information Processing over Networks
- Foundations of Data Science
- Qualitative Theory of Dynamical Systems

Provided faculty feedback on *Mathematics for Economics and Finance* (2nd Edition), Cambridge University Press, May 2025 — for potential adoption in undergraduate economics courses.

TECHNICAL SKILLS

- Programming: Python, MATLAB, R, SQL
- Mathematical and Machine Learning Methods: Topological data analysis, manifold learning, geometric representation learning, computer vision, statistical data analysis
- Scientific Computing and Research Tools: PyTorch, Jupyter, Git/GitHub, LaTeX, SPSS
- AI-Assisted Research: AI-assisted coding, literature review, and computational workflows using Claude and Codex

PROFESSIONAL ORGANIZATION MEMBERS

Member, Phi Kappa Phi Honor Society

Member, American Mathematical Society (AMS).

Member, International Linear Algebra Society (ILAS).

CONFERENCES/WORKSHOP ATTENDED

1. Organizer, AMS Special Session “Mathematical and Computational Methods for Biological Structures and Data,” Spring Southeastern Sectional Meeting, University of Alabama at Birmingham, Birmingham, AL, March 21, 2027 (upcoming).
2. Co-organizer, Special Session, Joint Mathematics Meetings (JMM 2027), Chicago, IL, January 12–15, 2027 (upcoming).
3. Organizer, MS34 “Mathematical and Computational Methods for High-Dimensional Biological Data,” SIAM Texas-Louisiana Sectional Meeting, Southern Methodist University, Dallas, TX, October 30–November 1, 2026 (upcoming).
4. Workshop on Molecular Biophysical Approaches for Drug Discovery Led by Prof. Duc Nguyen, Michigan State University, March 9–10, 2026
5. 2025 SIAM Great Lakes Section Annual Meeting, Chicago, September 27–28, 2025.
6. Volunteer, The Association of Mathematics Teacher Educators (AMTE) 2025 Annual Conference, Grand Sierra Resort, Reno, NV, February 6–8, 2025.
7. Chair, Special Session on Matrix Analysis and Its Applications, Joint Mathematics Meetings (JMM 2025), Seattle, January 8–11, 2025.

8. The International Conference “Recent Progress in Operator Theory and Its Applications”, Research Institute for Mathematical Sciences (RIMS), Kyoto University, Japan, November 6–8, 2024 (Virtual).
9. The 2024 Workshop Matrices and Operators (MAO 2024), Reno, USA, June 14-17, 2024.
10. Western Canada Linear Algebra Meeting (WCLAM), Calgary, Canada, May 25-26, 2024 (Virtual).
11. 2024 NSHE Corequisite Conference, Carson City, USA, April 25-26, 2024.
12. Joint Mathematics Meetings (JMM 2024), San Francisco, January 3-6, 2024.
13. The 10th International Conference on Matrix Analysis and Applications (ICMAA), KunMing, China, August 15-18, 2023 (Virtual).
14. International Workshop on Matrix Analysis and Its Applications, Quynhon, Vietnam, July 7-8, 2023 (Virtual).
15. Joint Mathematics Meetings (JMM 2023), Boston, January 4-7, 2023.
16. International Conference on Matrix Theory with Applications (ICMTA), Jeju, Korea, December 1-5, 2022 (Virtual).
17. Second International Workshop on Matrix Theory and Applications, virtual, November 15–17, 2022.
18. International Workshop on Matrix Analysis and Its Applications (MAA 2022), virtual, June 4, 2022.
19. Joint Mathematics Meetings (JMM 2022), virtual, April 6–9, 2022.
20. First NU-SIBAU International Workshop on Matrix Theory and Its Applications, virtual, October 15–17, 2021.
21. 2021 China-Korea International Conference on Matrix Theory with Applications (MIME 2021), virtual, November 26–27, 2021.